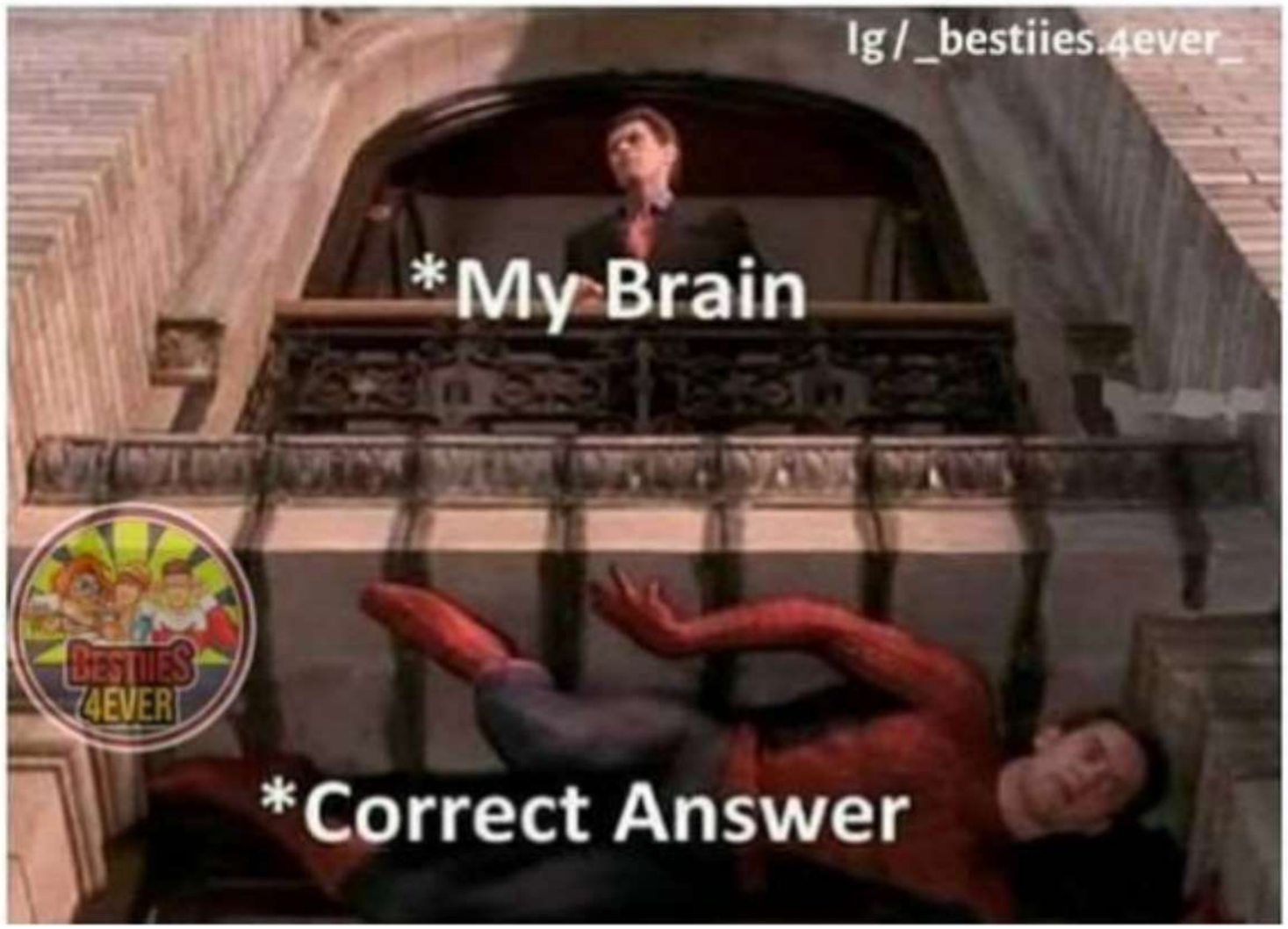
During exams



Chronic Inflammation

MDP10208 Man in the Environment
2025/2026

Prof. Dr. Dayangku Norlida A.O

Specific Instructional Objectives

At the end of the teaching session, the student should be able to:

1. describe the cells that are involved in chronic inflammation.
2. describe the general characteristics of chronic inflammation.
3. list examples of chronic inflammatory condition.
4. describe features of chronic granulomatous inflammation and their characteristic morphologic features.

Chronic Inflammation

a. Causes of Chronic Inflammation
b. Morphologic Features
c. Cells and Mediators of Chronic Inflammation
d. Granulomatous Inflammation

Chronic Inflammation

What is Chronic Inflammation?

- Chronic inflammation is a prolong inflammation that continues for weeks or months.
- It happens when the body cannot fully remove the injurious agent or when the injury keeps happening.

Chronic inflammation has three processes happening at the same time:

1. Ongoing inflammation – the immune system keeps responding.
2. Tissue injury – the prolonged response causes damage to normal tissue.
3. Attempts at repair – the body tries to heal by forming new blood vessels and fibrosis (scarring).

How Does It Start?

- Chronic inflammation can begin in two ways:

1. **After acute inflammation**
   - When the acute response fails to eliminate the cause (e.g., persistent infection).
   - *Example:* unresolved pneumonia progressing to chronic lung inflammation.

2. **Starts slowly without acute inflammation**
   - A low-grade, silent onset with no obvious acute phase.
   - *Example:* tuberculosis, autoimmune diseases.

Chronic Inflammation: (a) Causes of Chronic Inflammation

When Does Chronic Inflammation Occur?

- Chronic inflammation happens when the body cannot remove the cause of injury or when the immune system reacts repeatedly.

- There are three main settings:

1. Persistent Infections

- These are infections that are difficult to eliminate.

  Example: Mycobacteria, certain viruses, fungi, and parasites.
- They often trigger delayed-type hypersensitivity.
- May form granulomas — a special form of chronic inflammation.
- An acute infection may progress to chronic inflammation if unresolved
  Example: unresolved pneumonia → chronic lung abscess.
- Acute and chronic inflammation may occur together
  Example: peptic ulcer (acute ulcer + chronic gastritis).

Chronic Inflammation: (a) Causes of Chronic Inflammation

(2) Hypersensitivity diseases.

- These are excessive or abnormal immune reactions.

  a) Autoimmune diseases

  Caused by: - genetic predisposition - hormones and sex
  - environmental trigger - multiple autoimmune tendency
  (Epstein-Barr virus)

  - The immune system attacks self-tissues → continuous inflammation.
  - Examples: rheumatoid arthritis, multiple sclerosis.

b) Excessive reactions against microbes

- The immune system becomes overactive even to normal gut flora.
- *Example:* inflammatory bowel disease (IBD).

  - Crohn's disease
  - ulcerative colitis

c) Allergic reactions

- Immune response to harmless environmental antigens.
- *Example*: bronchial asthma.
- These conditions often show repeated episodes of inflammation.
- Over time, fibrosis (scarring) becomes dominant.

Chronic Inflammation: (a) Causes of Chronic Inflammation

(3) Prolonged exposure to potentially toxic agents

a) Exogenous (from outside the body)

- Silica dust → cannot be degraded → chronic lung inflammation (silicosis).

b) Endogenous (from within the body)

- Excess cholesterol/lipids deposited in blood vessels →
- ongoing inflammation → atherosclerosis.

Chronic Inflammation: (b) Morphologic Features

Characteristics of Chronic Inflammation

- Acute inflammation = quick onset, swelling, vascular changes, and neutrophils.
- Chronic inflammation = long-lasting with different cell types and healing attempts.

Chronic inflammation is characterised by three main features:

1. **Infiltration with Mononuclear Cells**

- These are long-living immune cells that dominate chronic inflammation:
  - Macrophages
  - Lymphocytes
  - Plasma cells
- They replace the short-lived neutrophils seen in acute inflammation.

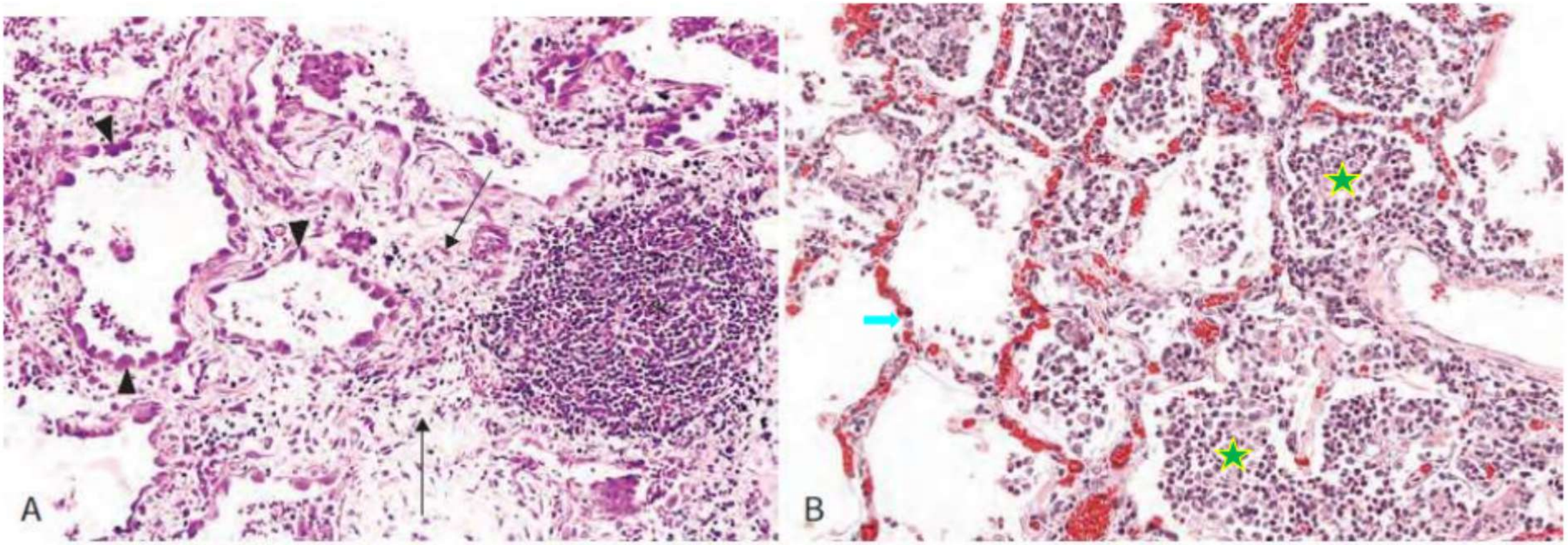
2. Ongoing Tissue Destruction

- Occurs because the injurious agent persists, OR
- Because activated immune cells cause damage while trying to eliminate the cause.

3. Attempts at Healing

- The body tries to repair the damaged area through:
  - Angiogenesis → growth of new small blood vessels
  - Fibrosis → collagen deposition leading to scar formation
- Healing and damage happen simultaneously.

Chronic Inflammation: (b) Morphologic Features



(A) Chronic inflammation in the lung, showing all three characteristic histologic features: (1) collection of chronic inflammatory cells (asterisk), (2) destruction of parenchyma (normal alveoli are replaced by spaces lined by cuboidal epithelium) (arrowheads) and (3) replacement by connective tissue (fibrosis) (arrows).

by inflammatory cells (produce protease, ROS, cytokine)

form type II pneumocyte hyperplasia & bronchiolization

(B) In contrast, in acute inflammation of the lung (acute bronchopneumonia), neutrophils fill the alveolar spaces ★ and blood vessels are congested. ➡

Chronic Inflammation: (c) Cells and Mediators of Chronic Inflammation

Role of Macrophages in Chronic Inflammation

- Macrophages are the key cells in chronic inflammation. They stay longer at the site of injury and perform several important functions.

Why Macrophages Are Important

- Macrophages dominate chronic inflammation because they:

1. Secrete Mediators

- They release:
  - Cytokines
  - Growth factors
- These substances:
  - Attract more immune cells
  - Activate other cells, especially T lymphocytes
  - Maintain and amplify inflammation

2. Destroy Harmful Agents

- Macrophages are professional phagocytes.
- They:
  - phagocytose and remove microbes
  - Clear dead or damaged tissue

3. Help in Tissue Repair

- Macrophages also:
  - Promote wound healing
  - Stimulate angiogenesis (new vessels)
  - Promote fibrosis (scar formation) in chronic inflammation

Chronic Inflammation: (c) Cells and Mediators of Chronic Inflammation

Where Do Macrophages Come From?

- Macrophages originate from two main sources:
- Postnatal life:
  - From haematopoietic stem cells in the bone marrow → these become monocytes in the blood → then enter tissues and become macrophages.
- Early development:
  - From yolk sac and foetal liver progenitors → these form long-living tissue macrophages present from birth.

What Are Monocytes?

- Monocytes = circulating precursors of macrophages.
- They travel in the blood and enter tissues when needed.

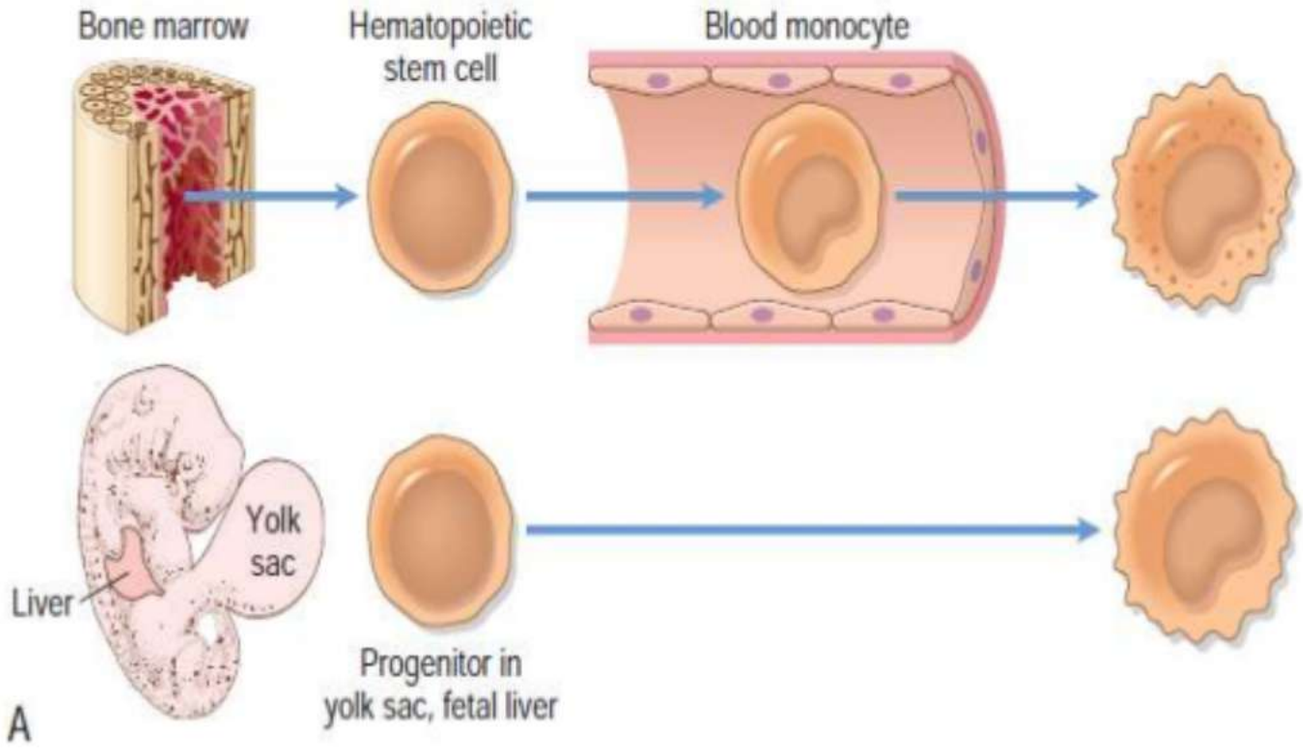
Where Are Macrophages Found?

- Macrophages are located almost everywhere in the body, especially in connective tissues.
- In some organs, they have special names:
  - Liver – Kupffer cells
  - Spleen & Lymph nodes – sinus histiocytes
  - Central nervous system – Microglial cells
  - Lungs – alveolar macrophages

What System Do They Form?

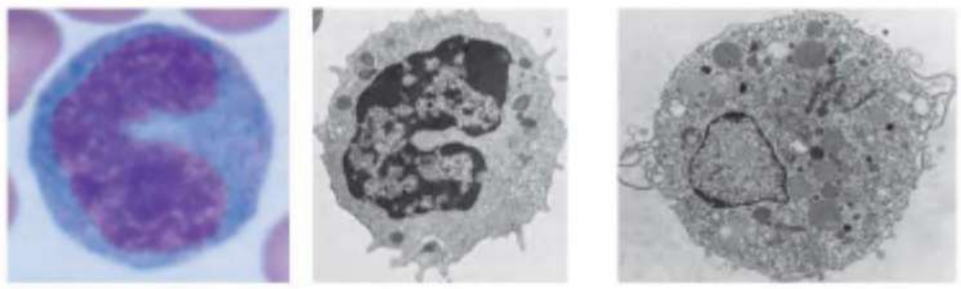
- All macrophages + monocytes together form the:
  - **Mononuclear Phagocyte System**
- (Previously called the *reticuloendothelial system*, but this term is outdated and inaccurate.)

Maturation of mononuclear phagocytes.



Origin of Macrophages
After birth: From bone marrow → monocytes → tissue macrophages (increase during injury/inflammation).

Before birth: Some (e.g., microglia, alveolar macrophages) come from embryonic precursors and stay for life.



(B) The morphology of a monocyte and activated macrophage

Chronic Inflammation: (c) Cells and Mediators of Chronic Inflammation

Macrophage Development

1. Bone Marrow → Monocytes → Macrophages

- Bone marrow progenitors produce monocytes that migrate into tissues and become macrophages, especially in skin and the intestinal tract.

2. Lifespan Differences

- Monocytes: short lifespan (~1 day in blood)
- Tissue macrophages: long lifespan (months to years)

3. Embryonic-Origin Tissue Macrophages

- Some specialised macrophages form before birth and remain lifelong:
  - Microglia (brain)
  - Kupffer cells (liver)
  - Alveolar macrophages (lungs)

4. Monocytes in Early Inflammation

- Monocytes enter inflamed tissues early.
- By 48 hours, they often become the dominant cell.
- Their movement into tissues uses the same adhesion and chemotactic mechanisms as neutrophils.

Chronic Inflammation: (c) Cells and Mediators of Chronic Inflammation

Functions of activated macrophages in chronic inflammation

- Activated macrophages have dual roles — they help protect the body but can also cause tissue damage.

1. Eliminate harmful agents

- They ingest and destroy microbes and dead tissue (phagocytosis).

2. Start and maintain inflammation

- Macrophages release inflammatory mediators such as:
  - Cytokines: TNF, IL-1
  - Chemokines
  - Eicosanoids
- These substances initiate and amplify inflammation.

3. Cause tissue injury

- The same mediators that kill microbes can also damage normal tissue, contributing to chronic inflammation.

4. Promote healing and fibrosis

- Macrophages stimulate:
  - Tissue repair
  - Scar formation
  - Fibrosis

5. Interact with T lymphocytes

- They present antigens to T cells.
- They also respond to T-cell signals, creating a feedback loop important in cell-mediated immunity.

Chronic Inflammation: (c) Cells and Mediators of Chronic Inflammation

Macrophage Activation Pathways

- Macrophages can be activated in two main ways:
  1. Classical pathway (M1)
  2. Alternative pathway (M2)

Classical (M1) Macrophage Activation

How M1 macrophages are activated

- M1 activation is triggered by:
  - Microbial products (e.g., endotoxin via TLRs) toll like - receptor
  - IFN-γ from T cells (Interferon)
  - Foreign particles (e.g., crystals, particulate matter)

What M1 macrophages do

- M1 macrophages:
  - Produce nitric oxide (NO) and lysosomal enzymes → kill microbes
  - Release pro-inflammatory cytokines → increase inflammation

- Main role of M1 macrophages
  - Destroy microbes
  - Promote strong inflammatory responses

Chronic Inflammation: (c) Cells and Mediators of Chronic Inflammation

Alternative (M2) Macrophage Activation –
How M2 macrophages are activated

- Triggered by cytokines IL-4 and IL-13
- (from T lymphocytes and other cells).

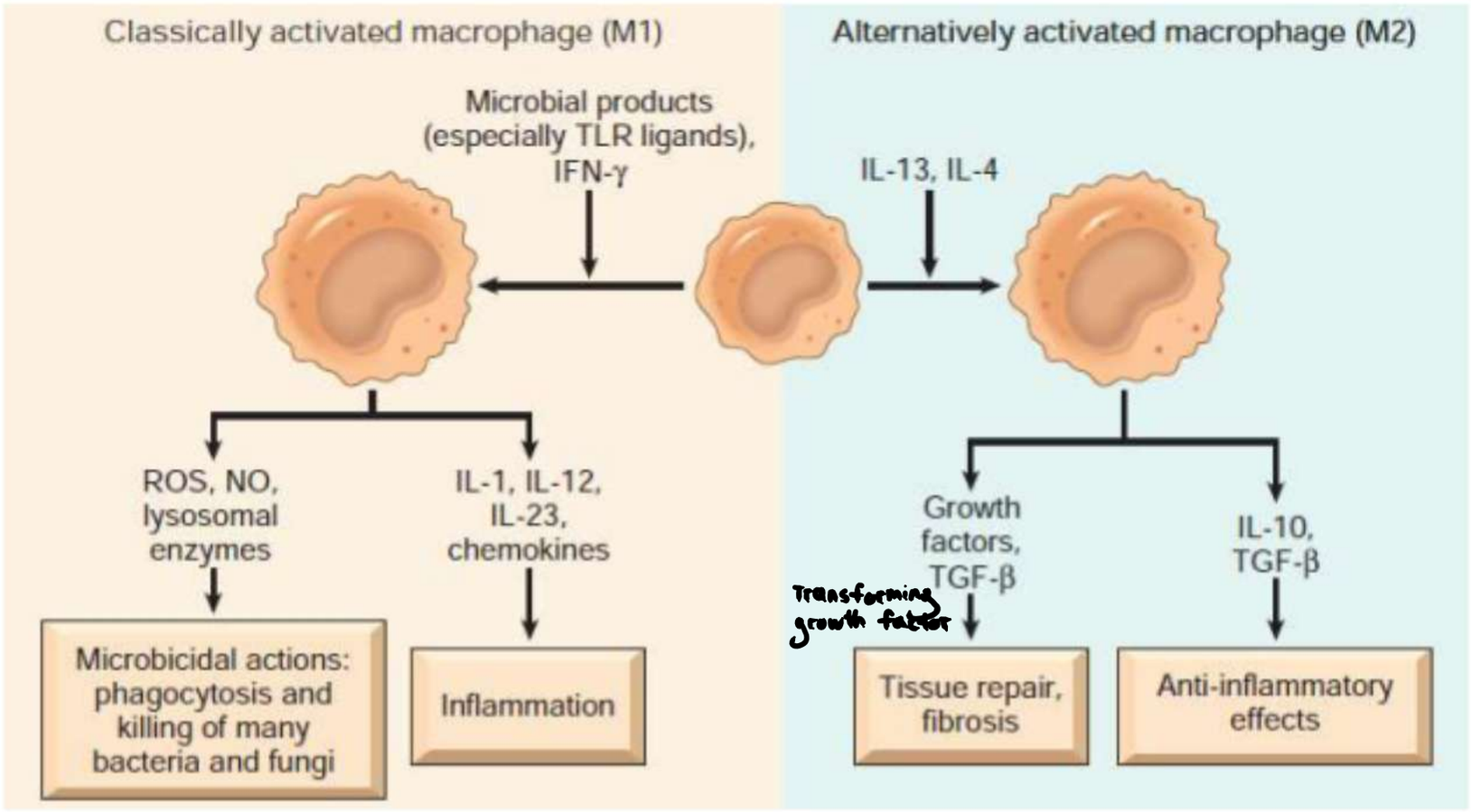
What M2 macrophages do

- M2 macrophages:
  - Reduce inflammation
  - Promote tissue repair and healing
  - Stimulate fibrosis when repair becomes excessive
- They are not effective at killing microbes.

Relationship with M1 pathway

- Usually, M1 activation comes first to kill microbes.
- M2 activation follows to help with repair.
- In reality, both types can appear together in most inflammatory reactions.

Classical and alternative macrophage activation



Classically activated macrophages are induced by microbial products and cytokines, particularly interferon-γ (IFN-γ). They phagocytose and destroy microbes and dead tissues and can potentiate inflammatory reactions.

Alternatively activated macrophages are induced by other cytokines and important in tissue repair and resolution of inflammation.

Chronic Inflammation: (c) Cells and Mediators of Chronic Inflammation

Macrophages and tissue destruction in chronic inflammation

1. Dual nature of macrophages

- Macrophages release many powerful mediators.
- These help defend the body, but when activation is excessive or prolonged, they also damage normal tissues.

➔This is why tissue destruction is a key feature of chronic inflammation.

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3. What happens to macrophages after the irritant is gone?

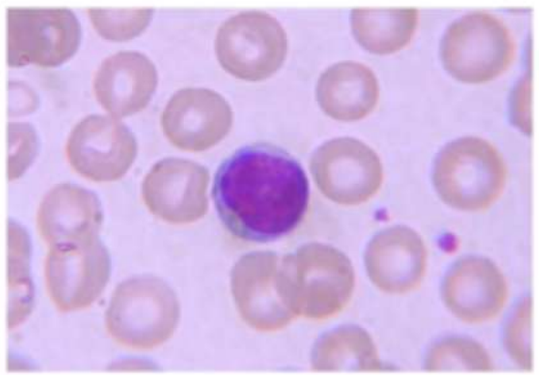
- If the harmful stimulus is removed → macrophages gradually disappear
- (they die or drain into lymphatics).

3. What happens if the irritant persists?

- Macrophages continue to accumulate because of:
  - Ongoing recruitment from the blood
  - Local proliferation at the site of inflammation
- This leads to persistent chronic inflammation and ongoing tissue damage

Chronic Inflammation: (c) Cells and Mediators of Chronic Inflammation

Role of Lymphocytes in Chronic Inflammation



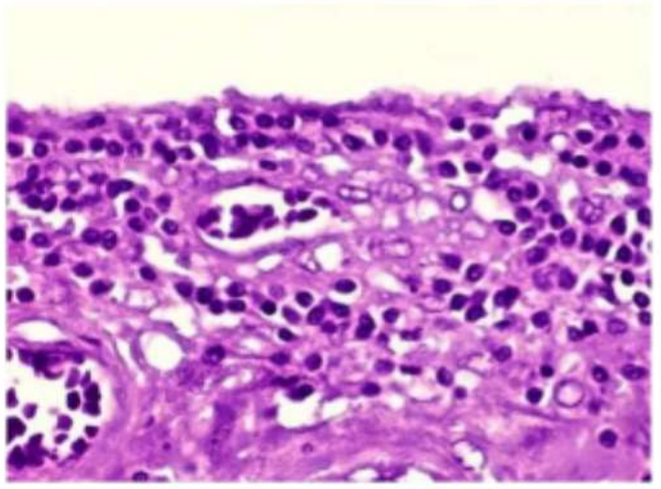
Lymphocytes in PBF

1. Activation by antigens

- Microbes and environmental antigens activate T cells and B cells.
- Once activated, lymphocytes amplify and maintain chronic inflammation.

2. Why they make inflammation persistent

- Lymphocyte activation creates long-lived memory cells.
- Because of these memory cells, inflammation becomes more prolonged and severe.



Lymphocytes in tissue

3. Importance in granulomatous inflammation

- Some forms of chronic inflammation, especially granulomas, rely heavily on memory T-cell responses.

Chronic Inflammation: (c) Cells and Mediators of Chronic Inflammation

cluster of differentiation (helper T-cell)

Role of CD4+ T Lymphocytes in Chronic Inflammation

- CD4+ T cells help control and shape inflammation because they release cytokines.
- There are three main subsets, each driving a different type of inflammatory response.

(when triggered by pathogen)

1. Th1 Cells

- Produce IFN-γ
- Activate macrophages via the classical (M1) pathway
- Important for fighting bacteria and viruses
- Also involved in autoimmune diseases

2. Th2 Cells

- Produce IL-4, IL-5, IL-13
- Recruit and activate eosinophils
- Promote alternative (M2) macrophage activation
- Important in helminth (worm) infections and allergic inflammation 七常七

3. Th17 Cells

- Produce IL-17 and related cytokines
- Stimulate chemokine production → recruit neutrophils and monocytes
- Important in defense against extracellular bacteria and fungi
- Contribute to inflammation in autoimmune diseases

Chronic Inflammation: (c) Cells and Mediators of Chronic Inflammation

Lymphocyte–macrophage interaction in chronic inflammation

1. Two-way communication

- Lymphocytes and macrophages activate each other, creating a loop that keeps chronic inflammation going.

2. What macrophages do

- **Macrophages:**
  - Present antigens to T cells
  - Provide co-stimulatory signals
  - Release cytokines (e.g., IL-12) that activate T cells

3. What activated T cells do

- **Activated T cells release cytokines** that:
  - Recruit more macrophages
  - Activate macrophages further
  - Lead to even more antigen presentation and inflammatory mediator release

4. Result

- This forms a self-sustaining cycle that prolongs inflammation.
- A common outcome of this persistent T-cell–macrophage interaction is the formation of granulomas.

Chronic Inflammation: (c) Cells and Mediators of Chronic Inflammation

Role of B lymphocytes & plasma cells in chronic inflammation

1. Presence of B cells and plasma cells

- Activated B cells and plasma cells are commonly found at sites of chronic inflammation.

2. Antibody production

- The antibodies produced may target:
  - Persistent microbes
  - Self-antigens (autoimmune)
  - Damaged tissue components
- Their exact role in many chronic inflammatory diseases is not fully understood.

3. Formation of tertiary lymphoid organs

- In some chronic inflammatory conditions, lymphocytes and plasma cells cluster together to form structures similar to lymph node follicles.
- These are called tertiary lymphoid organs.
- Examples seen in:
  - Rheumatoid arthritis (synovium)
  - Hashimoto thyroiditis (thyroid)

autoimmune condition

Why this matters

- These organised follicles may sustain and amplify the local immune reaction

Chronic Inflammation: (c) Cells and Mediators of Chronic Inflammation

Eosinophils in Chronic Inflammation (Simplified)

1. When eosinophils appear

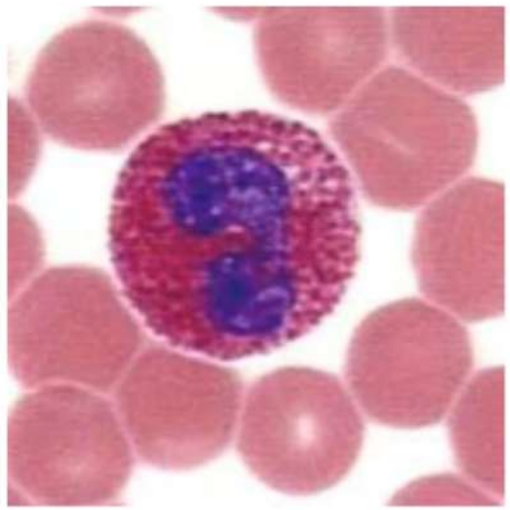
- Eosinophils are common in:
  - IgE-mediated immune reactions (e.g., allergies)
  - Parasitic infections (especially helminths)

2. How they are recruited

- They are attracted by:
  - The same adhesion molecules used by neutrophils
  - Eotaxin, a chemokine released by leukocytes and epithelial cells

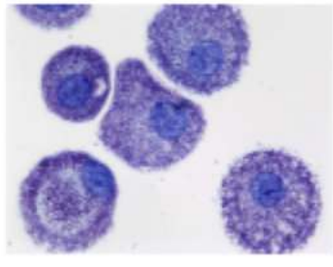
3. What they do

- Eosinophils contain major basic protein:
- Toxic to helminths → helps fight parasites
- Can also damage host epithelial cells → contributes to allergic tissue injury



Eosinophils: two lobes to their nucleus. These cells have large acidophilic specific granules - these stain bright red, or reddish-purple. These granules contain proteins that are 'destructive' and toxic.

Chronic Inflammation: (c) Cells and Mediators of Chronic Inflammation



Mast cells in chronic inflammation

1. Where they are

- Mast cells are found throughout connective tissues.
- They participate in both acute and chronic inflammation.

2. How they are activated

- They have FcεRI receptors that bind IgE antibodies.
- When these IgE-coated mast cells encounter their specific antigen → degranulation occurs.

3. What they release

- Histamine, prostaglandins, and other mediators.
- These trigger immediate hypersensitivity reactions, including allergies to:
  - Food
  - Insect venom
  - Drugs

  → Can lead to severe anaphylactic shock.

4. Role in chronic inflammation

- Mast cells also release many cytokines.
- These cytokines help sustain and amplify chronic inflammation.

Chronic Inflammation: (c) Cells and Mediators of Chronic Inflammation

Neutrophils in Chronic Inflammation



1. Neutrophils are not only for acute inflammation

- Although neutrophils mainly appear in acute inflammation, they can also be present in chronic inflammation.

2. Why neutrophils persist

- Large numbers of neutrophils may remain because of:
  - Persistent microbes
  - Cytokines released by activated macrophages and T cells

Neutrophil:
There is a single nucleus, which is multilobed, and can have between 2-5 lobes.

3. Examples

骨髓炎

- Chronic bone infection (osteomyelitis): neutrophils may persist for months
- Chronic lung damage from smoking or irritant exposure: neutrophils contribute to ongoing injury

4. "Acute on chronic" inflammation

- This pattern refers to chronic inflammation with repeated or ongoing acute neutrophilic episodes.



(d) Granulomatous Inflammation

Chronic Inflammation: (d) Granulomatous Inflammation

Granulomatous Inflammation

1. What it is

- A special type of chronic inflammation made up of:
  - Clusters of activated macrophages
  - Surrounding T lymphocytes
  - Sometimes central necrosis

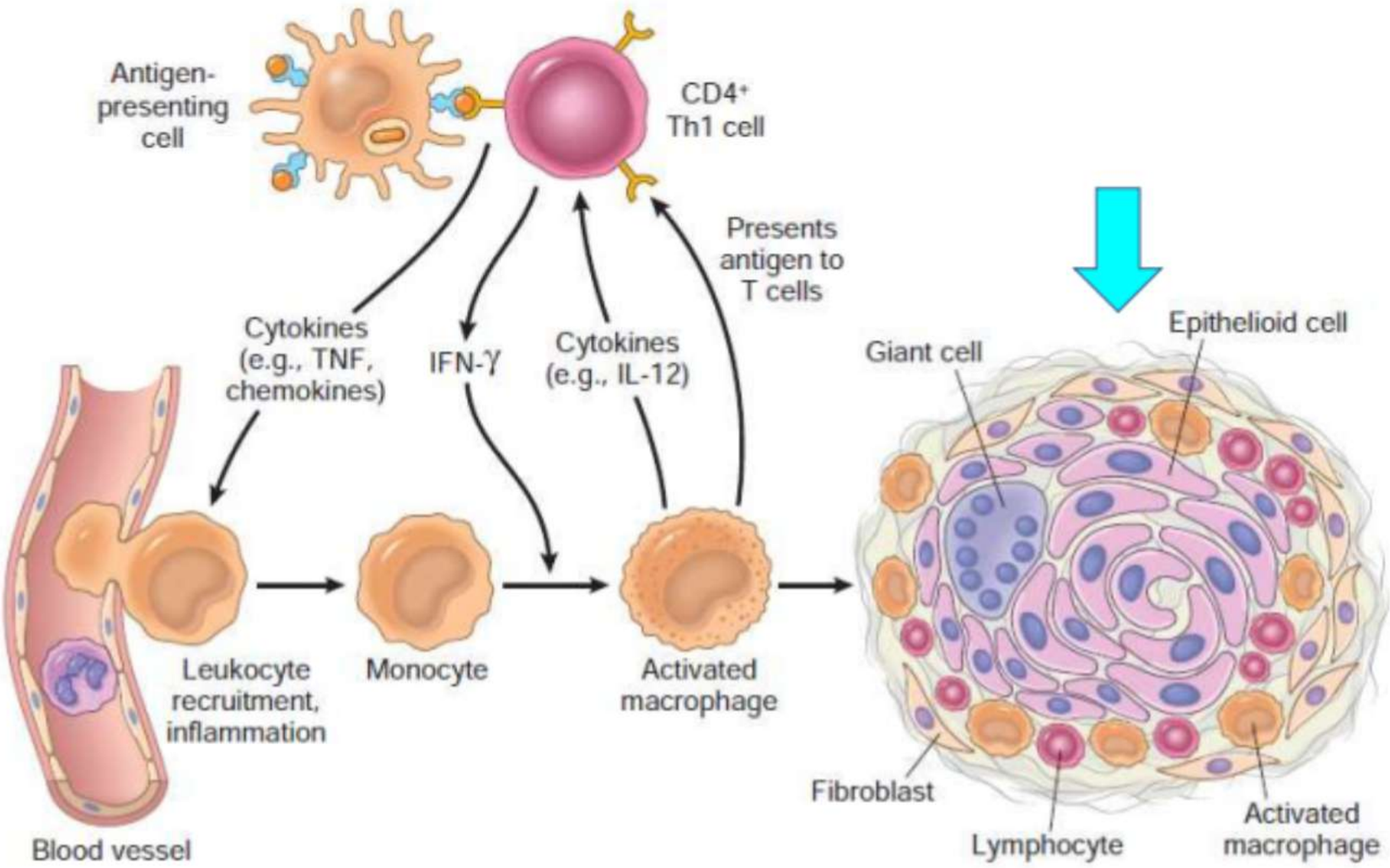
2. Why granulomas form

- Granulomas form when the body tries to contain an agent that is hard to eliminate
  - (e.g., certain bacteria, fungi, foreign bodies).
- This involves strong T-cell activation, which further activates macrophages.

3. What the macrophages become

- Activated macrophages develop abundant pink cytoplasm and look like epithelial cells → called **epithelioid cells.**
- Some macrophages fuse together to form **multinucleated giant cells.**

Granuloma



Macrophage–Lymphocyte Interactions in Chronic Inflammation

1. What T cells do

- Activated T cells release cytokines that:
  - Recruit macrophages → TNF, IL-17, chemokines
  - Activate macrophages → IFN-γ

2. What macrophages do

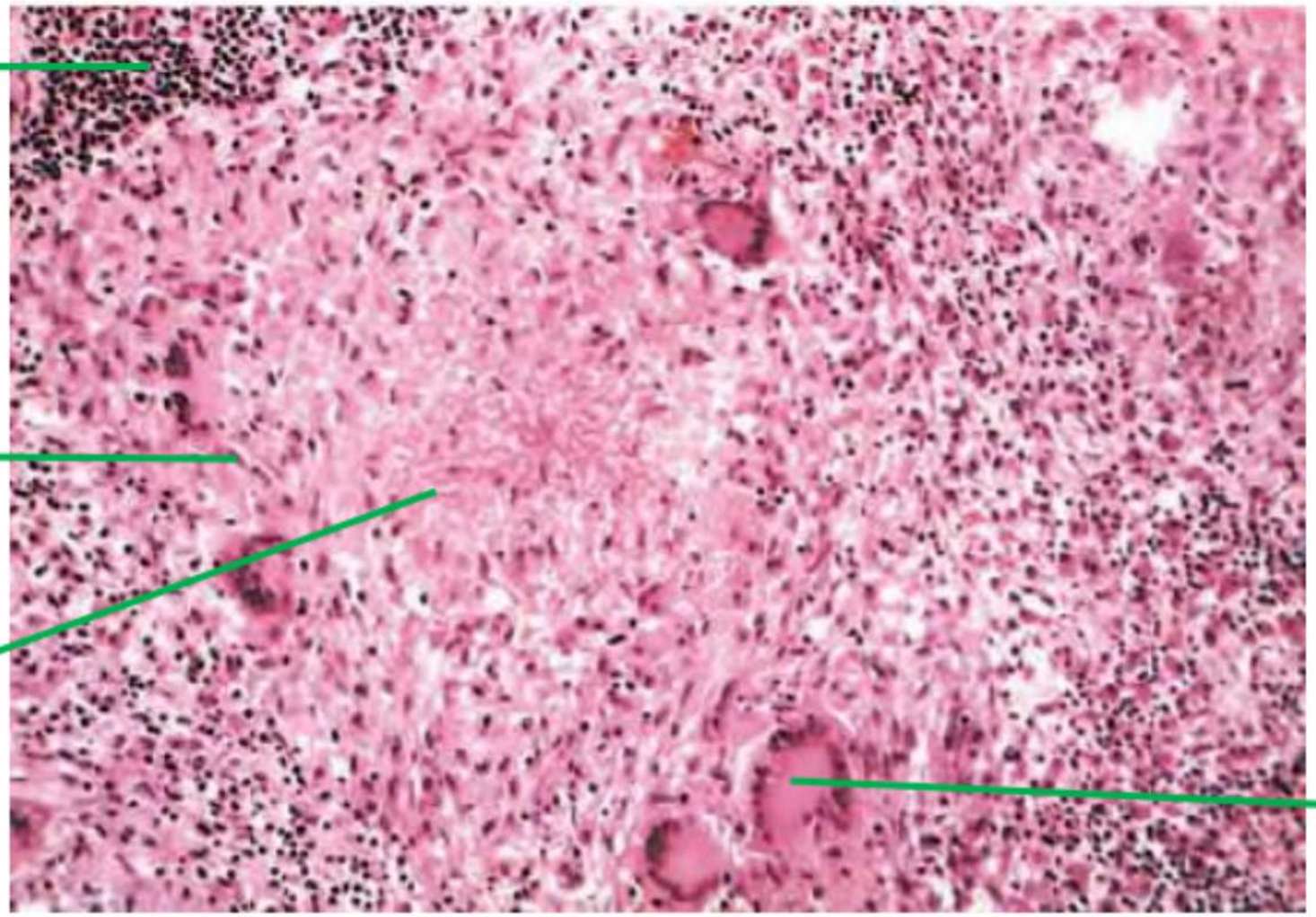
- Activated macrophages:
- Present antigens to T cells
- Release cytokines such as IL-12 → further activate T cells

3. Result of this cycle

- This two-way activation creates a persistent inflammatory loop.
- If it continues for a long time → granulomas may form.

Granuloma

Granuloma: aggregates of activated macrophages (epithelioid histiocytes) collared by lymphocytes (often T cells) and sometimes associated with central necrosis and multinucleated Langhans-type giant cells.



Typical tuberculous granuloma showing an area of central necrosis surrounded by multiple Langhans-type giant cells, epithelioid cells and lymphocytes.

multiple activated macrophage fuse tgt (seen in granulomas during chronic infection, eg. TB)

Chronic Inflammation: (d) Granulomatous Inflammation

Foreign Body Granulomas

1. What causes them

- Form when the body reacts to inert foreign materials
- (e.g., talc, sutures, or other fibres).

滑石

2. Immune response

- They occur without a strong T-cell–mediated immune reaction.
- The material is not immunogenic (does not trigger specific immunity).

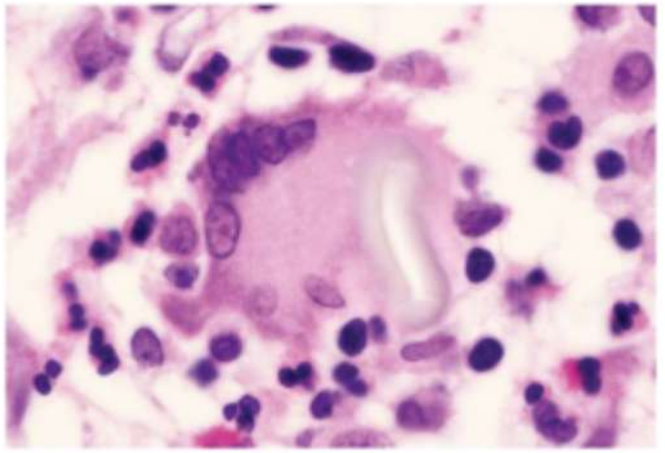
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3. Why granulomas form

- The foreign material is too large to be phagocytosed by macrophages.
- Macrophages become activated and form:
  - Epithelioid cells
  - Multinucleated giant cells (mngc)

4. Appearance

- Epithelioid cells and giant cells sit against the surface of the foreign body.
- The foreign material is usually visible in the centre of the granuloma, sometimes inside giant cells.



Mngc in foreign body granuloma that phagocytosed a suture material for wound suture

Chronic Inflammation: (d) Granulomatous Inflammation

Immune Granulomas

1. What causes them

- Form when an agent triggers a persistent T-cell–mediated immune response.
- Usually occur when the offending agent (e.g., persistent microbes) is difficult to eliminate.

2. Key immune mechanism

- Th1 cells release IFN-γ, which strongly activates macrophages.
- These activated macrophages then form epithelioid cells and giant cells, creating the granuloma.

3. Special case: Th2-driven granulomas

迴收告病

- In some parasitic infections (e.g., schistosomiasis), granulomas are driven by Th2 responses.
- These granulomas often contain many eosinophils.

Chronic Inflammation: (d) Granulomatous Inflammation

Clinical Importance of Granulomatous Inflammation

1. Why recognising granulomas matters

- Granulomas have a limited number of causes, so identifying them helps narrow down the diagnosis.

2. Tuberculosis (TB) — the most important cause

- TB is the classic (prototype) granulomatous disease.
- When granulomas are seen, TB must always be considered and excluded.
- A TB granuloma is called a tubercle.

3. Other causes of granulomatous inflammation

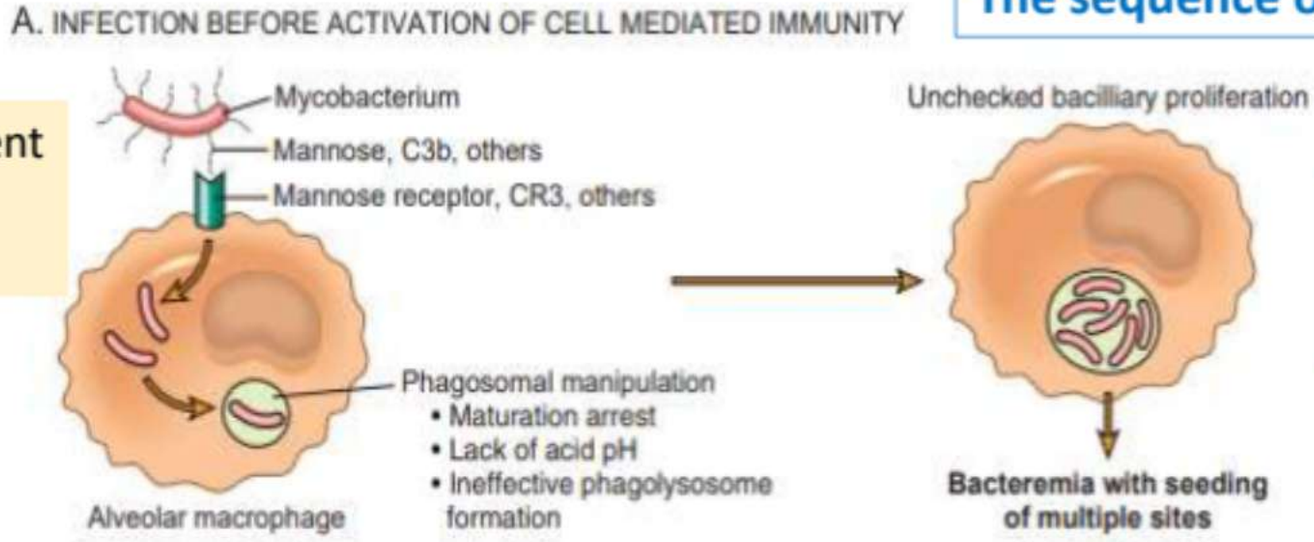
- Granulomas can also occur in:
  - Crohn disease → immune-mediated inflammatory bowel disease
  - Sarcoidosis → chronic systemic disease of unknown cause
  - Certain fungal infections
  - Syphilis and other chronic infections

4. How to identify the cause

- Because many conditions look similar under the microscope, special investigations are needed to confirm the cause:
  - Special stains
    - Acid-fast stain → TB
    - Fungal stains → fungal infections
  - Culture methods → TB, fungi
  - Molecular tests (PCR) → TB and others
  - Serologic tests → syphilis

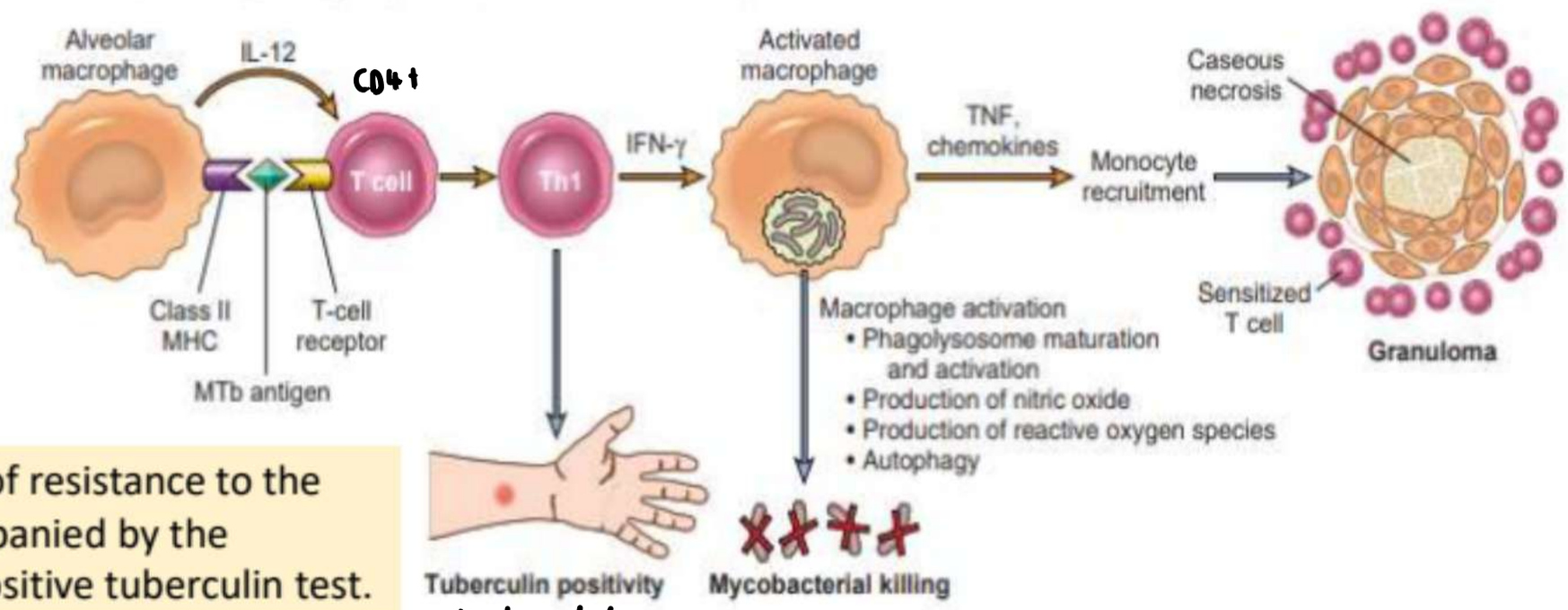
The sequence of events in primary pulmonary TB

Inhalation of virulent *Mycobacterium tuberculosis*



Development of T cell-mediated immunity to the organism.

The initiation and consequences of T-cell–mediated immunity.



Formation of granuloma

The development of resistance to the organism is accompanied by the appearance of a positive tuberculin test.

Causes of Granulomatous Inflammation!!!

Disease	Cause	Tissue Reaction
Tuberculosis	*Mycobacterium tuberculosis*	Caseating granuloma (tubercle): focus of activated macrophages (epithelioid cells), rimmed by fibroblasts, lymphocytes, histiocytes, occasional Langhans giant cells; central necrosis with amorphous granular debris; acid-fast bacilli
Leprosy	*Mycobacterium leprae*	Acid-fast in macrophages; non-caseating granulomas
Syphilis	*Treponema pallidum*	Gumma: microscopic to grossly visible lesion, enclosing wall of histiocytes; plasma cells infiltrate; central cells are necrotic without loss of cellular outline
Cat-scratch disease	Gram-negative bacillus	Rounded or stellate granuloma containing central granular debris and recognizable neutrophils; giant cells uncommon
Sarcoidosis	Unknown aetiology	Non-caseating granulomas with abundant activated macrophages
Crohn disease (inflammatory bowel disease)	Immune reaction against intestinal bacteria, possibly self-antigens	Occasional non-caseating granulomas in the wall of the intestines, with dense chronic inflammatory infiltrate

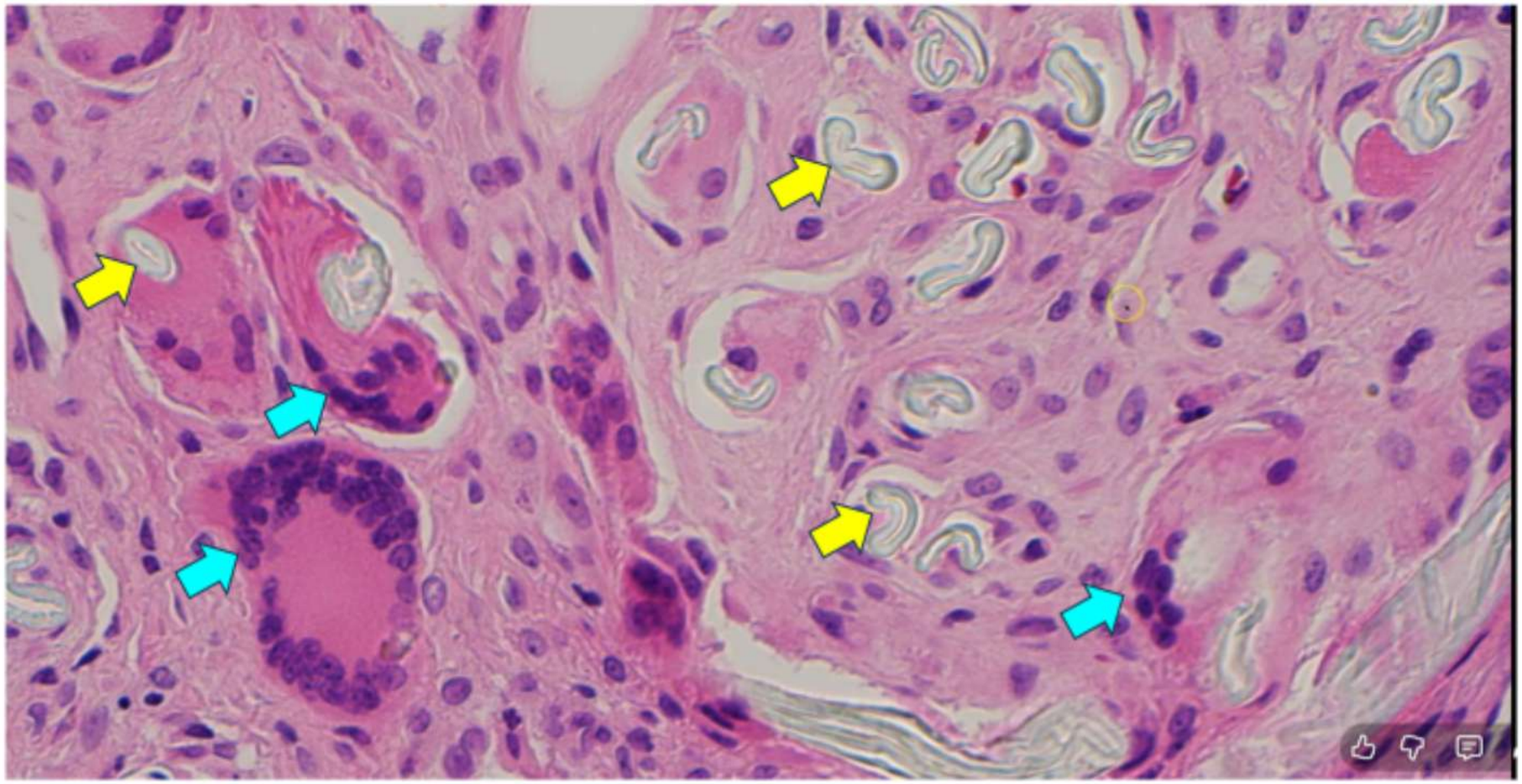


e) Foreign Body–Type Reaction

Foreign Body Granuloma

- A foreign body–type reaction occurs when non-living, non-degradable material enters tissues.
- Examples: sutures, talc, splinters, prosthetic material.
- These materials are too large or inert to be phagocytosed by macrophages.
- Macrophages become activated and may fuse to form multinucleated foreign body giant cells.
- Giant cells and epithelioid macrophages cluster around the surface of the foreign material.
- The reaction is not T-cell mediated (unlike immune granulomas).→ It occurs without a specific immune response.
- The goal is to wall off and isolate the foreign material rather than destroy it.
- Foreign body reactions are part of the chronic inflammatory response and may coexist with fibrosis and granulation tissue.

Foreign Body Granuloma



Foreign body – gauze (yellow arrow)
Multinucleated giant cell (blue arrow)

Systemic Effects Of Inflammation

Systemic Effects of Inflammation

Acute-Phase Response

1. What it is

- Even when inflammation is local, the body shows systemic (whole-body) changes caused by cytokines.
- These changes together are called the acute-phase response.

2. What it feels like

- Common symptoms during infections like influenza are examples of the acute-phase response:
  - Fever
  - Tiredness
  - Loss of appetite
  - Body aches

3. What causes it

- The reaction is driven by cytokines released during inflammation, especially:
  - TNF
  - IL-1
  - IL-6
  - Other cytokines (e.g., interferons) also play a role.

Systemic Effects of Inflammation

(1) Fever

What fever is

- A rise in body temperature (usually +1°C to +4°C)
- A common feature of the acute-phase response, especially during infections

What causes fever?

- Substances called pyrogens:

1. Exogenous pyrogens (from microbes)

- e.g. LPS from bacteria
- Stimulate immune cells to release IL-1 and TNF

2. Endogenous pyrogens (from the body)

- Mainly IL-1 and TNF

How fever happens

- IL-1 and TNF increase cyclooxygenase activity
- This ↑ production of prostaglandins (especially PGE2)
- PGE2 acts on the hypothalamus
- The hypothalamus resets body temperature to a higher level
  - ↓ heat loss (vasoconstriction)
  - ↑ heat production (brown fat & muscles)

How fever is reduced

- NSAIDs (e.g., aspirin) reduce fever by inhibiting prostaglandin synthesis

Systemic Effects of Inflammation

(2) Acute-Phase Proteins

What they are

- During inflammation, the liver produces more plasma proteins called acute-phase proteins.
- Their levels can rise hundreds-fold.

The three major acute-phase proteins are:

1. C-reactive protein (CRP)
2. Fibrinogen
3. Serum amyloid A (SAA)

What stimulates their production

- Cytokines from inflammation:
  - IL-6 → ↑ CRP & fibrinogen
  - IL-1 & TNF → ↑ SAA

Functions

- **CRP & SAA**
  - Bind to microbes → act as opsonins
  - Help activate complement
  - Bind to released chromatin → help clear necrotic cells
- **Fibrinogen**
  - Causes red blood cells to form rouleaux (stacking)
  - These stacks settle faster → basis of erythrocyte sedimentation rate (ESR) → **ESR** is a simple test for inflammation

Clinical relevance

- Useful and protective in acute inflammation
- Chronic elevation, especially of SAA, can lead to secondary amyloidosis

Systemic Effects of Inflammation

(2) Acute-phase proteins

1. CRP and Heart disease

- High CRP levels in the blood may indicate a higher risk of heart attack.
- Reason: inflammation in atherosclerotic plaques can trigger thrombosis, leading to myocardial infarction.

铁调素.

2. Hepcidin and Anaemia of chronic inflammation

- Hepcidin (a liver protein) increases during inflammation.
- It reduces iron availability for red blood cell production.
- Over time → leads to anaemia of chronic inflammation.

3. Thrombopoietin and Thrombocytosis

- Thrombopoietin (stimulates platelet production) is also increased.
- This can cause high platelet counts → thrombocytosis, often seen in chronic inflammatory states.

Systemic Effects of Inflammation

(3) Leukocytosis

1. What it is

- Leukocytosis = increased white blood cell (WBC) count
- Very common in inflammation, especially bacterial infections

2. How high can the WBC go?

- Typical increase: 15,000–20,000 cells/mL
- Extreme cases: 40,000–100,000 cells/mL
  - These very high counts are called leukemoid reactions
  - They resemble leukemia but are reactive, not cancer

3. Why leukocytosis happens

- **Early phase**
  - Cytokines (TNF, IL-1) cause rapid release of neutrophils from bone marrow
  - Both mature and immature neutrophils appear in blood

  → This is called a *"shift to the left"*

- **Later phase (prolonged infection)**
  - Bone marrow increases production of new neutrophils
  - Driven by colony-stimulating factors (CSFs) from activated macrophages and marrow stromal cells

Systemic Effects of Inflammation

(3) Leukocytosis

1. Why the bone marrow increases WBC production

- To replace leukocytes that are being used up at the site of inflammation.

2. Neutrophilia

- Seen in most bacterial infections
  → High neutrophil count

3. Lymphocytosis

- Seen in many viral infections such as:
  - Infectious mononucleosis
  - Mumps 腮腺炎
  - German measles (rubella) 风疹

→ High lymphocyte count

4. Eosinophilia

- Seen in:
  - Allergic reactions
  - Helminth (worm) infections

→ High eosinophil count

白细胞减少症

5. Leukopenia

- Seen in some infections (e.g., typhoid fever, certain viruses, rickettsiae, some protozoa)
- Caused partly by sequestration of WBCs in tissues

→ Low total white cell count

多形性
gram-negative, pleomorphic bacteria that act as obligate intracellular parasites

Systemic EffectsoInflammation

(4) Other manifestation

1. Physiologic changes during fever

- Cytokines act on the brain and cause:
  - ↑ Pulse and ↑ blood pressure
  - ↓ Sweating (blood is redirected away from the skin to reduce heat loss)
  - Rigors (shivering)
  - Chills (seeking warmth)
  - Anorexia, sleepiness, and malaise 接物用

Sepsis and Septic Shock

2. What happens in severe bacterial infections

- In sepsis, large numbers of bacteria and their toxins enter the blood.
- This triggers massive release of cytokines, especially:
  - TNF
  - IL-1

3. Effects of very high cytokine levels

- Leads to serious, life-threatening complications:
  - Disseminated intravascular coagulation (DIC)
  - Hypotensive shock
  - Metabolic disturbances
    - Insulin resistance
    - Hyperglycemia 高血糖

4. Septic shock

- The combination of:
  - DIC
  - Shock
  - Metabolic abnormalities
- is called septic shock, a severe form of **systemic inflammatory response syndrome (SIRS)**.
- Septic shock is often fatal and represents the most dangerous consequence of uncontrolled inflammation.

Systemic Effects of Inflammation

Defective Inflammation

1. Why it matters

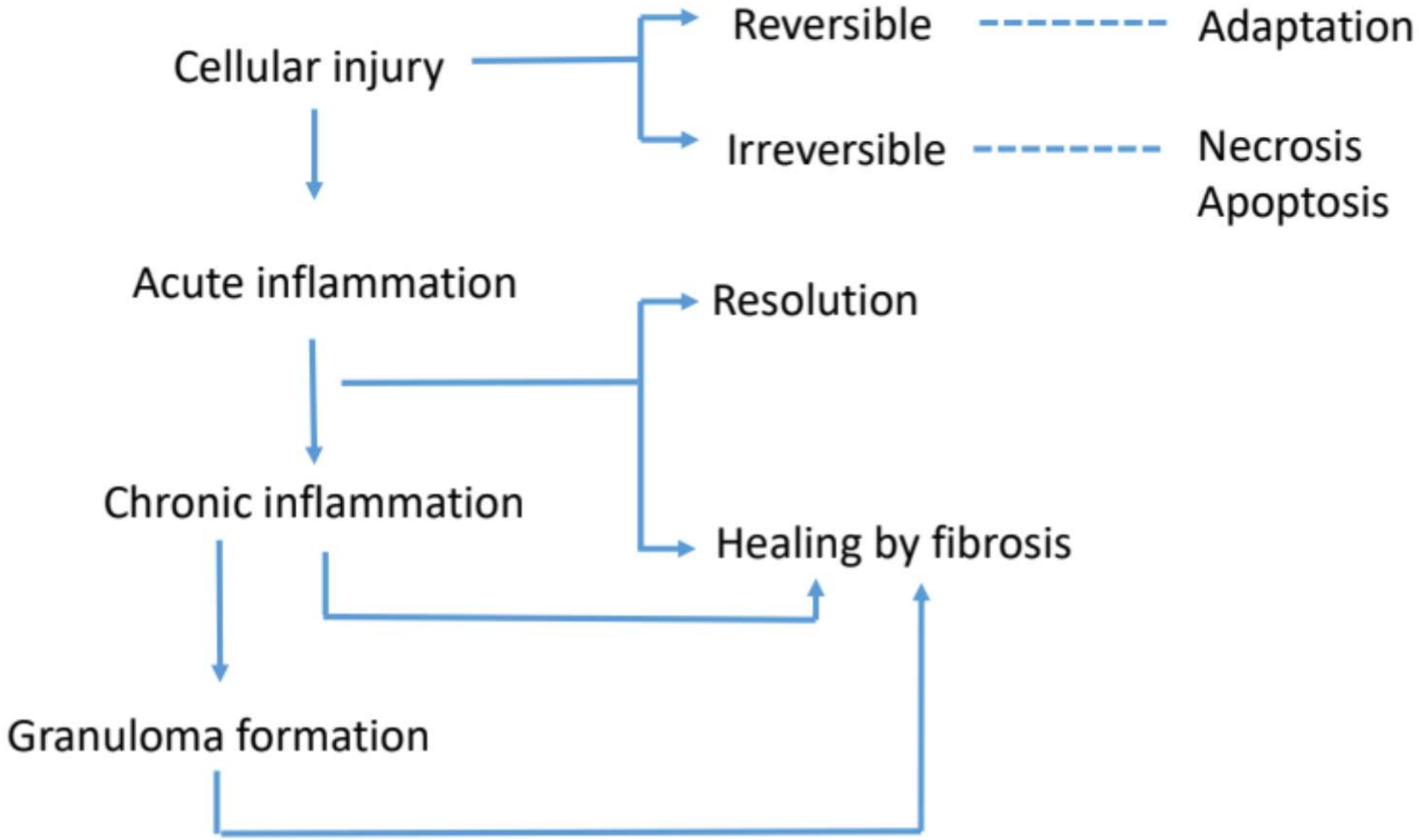
- Too much inflammation → many chronic diseases
- Too little inflammation → increased risk of infections

2. Most common cause

- Leukocyte deficiency
  - Often due to bone marrow replacement by:
    - Leukemia
    - Metastatic cancers
  - Also caused by marrow suppression from treatments for:
    - Cancer
    - Organ graft rejection

3. Rare causes

- Inherited defects in:
  - Leukocyte adhesion
  - Leukocyte killing (microbicidal) function
- These genetic disorders are uncommon.


Cellular injury
Reversible
Adaptation
Irreversible
Necrosis
Apoptosis
Acute inflammation
Resolution
Chronic inflammation
Healing by fibrosis
Granuloma formation

Reference

- Kumar, V., Abbas, A. K., Aster, J. C., Turner, J. R., & Perkins, J. A. (2021). Robbins & Cotran pathologic basis of disease. 10th Edn. Elsevier.

Thank you